

ASSESSMENT GUIDE

XPlatform

Specification and Design

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Glossary

AI	Artificial Intelligence
API	Application Programming Interface
CRM	Customer Relationship Management
CTR	Click-Through Rate
ER	Entity-Relationship (diagram)
LLM	Large Language Model
ML	Marketing-Qualified Lead
MRR	Monthly Recurring Revenue
OAuth	Open Authorization — a protocol for secure, delegated third-party access
REST	Representational State Transfer
SaaS	Software as a Service
SEO	Search Engine Optimisation
UI	User Interface

1. Executive Summary

XPlatform is a multi-tenant customer-intelligence and business-analytics platform built on Laravel 12, serving Master Admins (who manage the platform) and Clients (businesses who connect their website, email marketing, CRM, social media, chat support and payment tools into one place) [1]. This document specifies and designs the most recent development cycle: **Ask Mira**, a public homepage AI assistant that now performs real, free-tier website analysis and captures the visitor as a lead; three specialised **Business Helper AI agents** (Marketing, Sales, Customer Retention) that turn a client's real, synced CRM and email data into a ranked, step-by-step playbook rather than a static dashboard; a live **Brevo email-engagement panel** with recipient-level drill-down; and a live **HubSpot CRM analytics dashboard** [2]. The defining design decision across all four pieces is the same one: every number a client or visitor sees is computed from real, connected data — never a fixed demo figure — and every AI-written answer degrades gracefully to a correct, plain-text summary if no AI provider key is configured, rather than failing or fabricating an answer.

2. Specification

2.1 System Description

XPlatform consists of three portals sharing one Laravel 12 codebase and one multi-tenant database: a **public marketing site** (unauthenticated, includes the Ask Mira widget), a **Master Admin console** (`/admin/*`, client approvals and platform-wide settings), and a **Client application** (`/app/*`, the onboarding wizard, dashboards, and the Business Helpers workspace). A public tracking/collector API and an authenticated API sit alongside these for programmatic and JavaScript-tracker access [1].

The system described here is the layer that sits on top of a client's *already-connected* data sources. Once a client has connected their Brevo account (email marketing) and their HubSpot account (CRM) — using OAuth 2.0 connection flows that already existed in the platform before this cycle — that data is synced into XPlatform's own tables and three things now happen with it: it renders as live analytics panels inside XPlatform itself; it is scored and ranked into the Business Helper agents' recommendations; and, on the public side, Ask Mira uses the same real-data-first philosophy to turn an anonymous visitor into a scored lead before they are even a client.

2.2 Feasibility Analysis

2.2.1 Technical Feasibility

Technically feasible and largely de-risked, because every new capability was built as an additive layer on an already-proven stack rather than new infrastructure:

- **Framework & language** — Laravel 12 / PHP, matching the rest of the codebase; no new framework was introduced.

- **AI providers** — Ask Mira's general Q&A uses the Anthropic Claude API when a key is configured [3]; the Business Helper agents' written answers use the OpenAI API (`App\Services\Llm\OpenAiClient`) in the same request/response shape across all three agents. Both providers are optional at runtime — every agent method computes its numeric answer first with plain PHP arithmetic over the client's synced data, and only hands that computed answer to the LLM to phrase in plain English. If no key is configured, or the LLM call fails, the same correct, plain-text answer is returned instead of an error.
- **Third-party data** — Brevo and HubSpot both already had OAuth connection flows in the platform; this cycle reused them rather than building new ones (consistent with the Project Plan's assumption that "no rework needed for existing OAuth connection flows").
- **Background processing** — Brevo's API has no single endpoint that returns every delivered/opened/clicked/unsubscribed recipient, so that drill-down is built from a background sync job (`BrevoDeliveredRecipient`) with rate-limit-aware retry, rather than a synchronous request on every page view.

2.2.2 Economic Feasibility

The main new recurring cost is LLM API usage (Anthropic for Ask Mira, OpenAI for the Business Helper agents), billed per token and bounded by keeping each agent answer short (most prompts cap responses at 80–320 tokens) and by only calling the LLM once real data has already been computed — the LLM is never asked to do the scoring itself. Brevo and HubSpot API usage runs against the client's own connected account and existing plan, not a cost XPlatform bears directly. Against this, the business case from the Project Plan holds: converting more homepage visitors into qualified leads, and reducing the amount of time clients spend context-switching into Brevo/HubSpot's own dashboards, both have a direct, quantifiable value to the platform owner and to clients respectively.

2.2.3 Organizational Feasibility

The work matches the existing team structure defined in the Project Plan — Backend, Frontend, and AI/Integration streams, with a QA gate at the end of each feature stream [2]. No new roles or skills outside Laravel/PHP, front-end JavaScript, and LLM prompt-and-response integration were required; the AI-integration pattern (compute first, narrate second) was established once and repeated across every agent, keeping the learning curve for later features low.

2.3 Requirements Specification

2.3.1 Functional

Table 2.1 Functional Requirements

ID	REQUIREMENT	STATUS
FR1	A visitor can ask Ask Mira a general question about XPlatform (pricing, industries, product) and receive an answer, with or without an AI key	Implemented

	configured (local keyword fallback vs. Claude).	
FR2	A visitor can submit their own website's URL to Ask Mira and receive a real technical/SEO audit (0–100 score, letter grade, per-check breakdown) without signing up.	Implemented
FR3	Every Ask Mira URL analysis captures the visitor's name, email and submitted URL as a lead record.	Implemented
FR4	A logged-in client can connect their Brevo account and trigger a sync that pulls real delivered/opened/clicked/converted/unsubscribed counts.	Implemented
FR5	A client can drill down from any Brevo headline metric to the actual list of recipients behind it.	Implemented
FR6	A client can connect their HubSpot account and view a CRM analytics dashboard (Overview, Pipeline, Deals, Contacts, Activities, Forecast).	Implemented
FR7	The Marketing, Sales and Customer Retention Business Helper agents rank a client's real synced CRM contacts (blended with real Brevo engagement signal) into an actionable, prioritised list, replacing the previous fixed set of fictional demo accounts.	Implemented
FR8	Each Business Helper agent step (e.g. Marketing's Campaign/Performance/Audience/Insights/A/B Test, Sales' Prioritise/Understand/Pitch/Overcome/Close & grow, and Customer Retention's equivalent steps) answers its predefined quick-prompt questions by computing from real <code>crm_contacts</code> / <code>crm_deals</code> data first, then optionally phrasing that answer with an LLM.	Implemented for the large majority of prompts
FR9	Each agent panel has an embedded "Ask Mira" side-chat pre-loaded with agent- and step-specific quick prompts sourced from the database (<code>AgentPredefinedPrompt</code>), not hardcoded in the view.	Implemented
FR10	A small number of quick-prompt questions across the agents (see Section 2.3.1.1) are intentionally not yet backed by a real data source and return an honest "not available yet" message rather than a fabricated number.	Documented gap — see below

2.3.1.1 Honest scope note on FR8/FR10

Not every quick-prompt question across the three agents is backed by a real computed answer yet, and this specification records that honestly rather than overstating completeness. In the Customer Retention agent specifically: the Risk Radar step's "who to save first" and "who is drifting to the watchlist" questions are real (ranked from `crm_contacts` / `crm_deals` / email signal); three sibling questions in the same step (renewal risk, payment-failure risk, value-at-risk) return a plain, correctly-worded message stating that no renewal-date or payment-failure field is currently synced, rather than inventing a number for data the platform does not have. Similarly, the "Recover & Grow" step's six questions all depend on knowing the *outcome* of a previous save attempt (was the account saved, when did it

stabilise) — no table in the current schema records that outcome history yet, so this step also returns an honest placeholder message rather than fabricated figures. Marketing's and Sales' equivalent steps (Campaign, Performance, Audience, Insights, A/B Test, Manager for Marketing; Prioritise, Understand, Pitch, Overcome, Close & grow for Sales) were verified to follow the real-data-first pattern (each service class queries `CrmContact` / `CrmDeal` directly and only then, optionally, calls the LLM).

2.3.2 Non-functional

Table 2.2 Non-functional Requirements

ID	REQUIREMENT	CATEGORY
NFR1	No client data may be visible to another tenant, other than through aggregated views the Master Admin already had access to.	Security / Isolation
NFR2	Third-party tokens (Brevo, HubSpot OAuth access tokens) are stored encrypted at rest and never rendered back to the browser.	Security
NFR3	Every AI-narrated feature must still return a correct answer with no AI provider key configured — an LLM outage or missing key must never take an otherwise-working, data-grounded feature offline.	Reliability
NFR4	Brevo/HubSpot syncing must not call the third-party API synchronously on every page load; rate-limit-aware background jobs and caching are required.	Performance
NFR5	The Brevo panel must show the client exactly when data was last synced and from how many campaigns, so data freshness is never ambiguous.	Usability
NFR6	Every analytics report exposes a time-range selector and an export action, consistent with the rest of the platform's reporting screens.	Consistency
NFR7	New PHP code follows PSR-12 and existing Laravel conventions already used across the codebase.	Maintainability

2.4 Use Cases

2.4.1 Use Case Diagrams

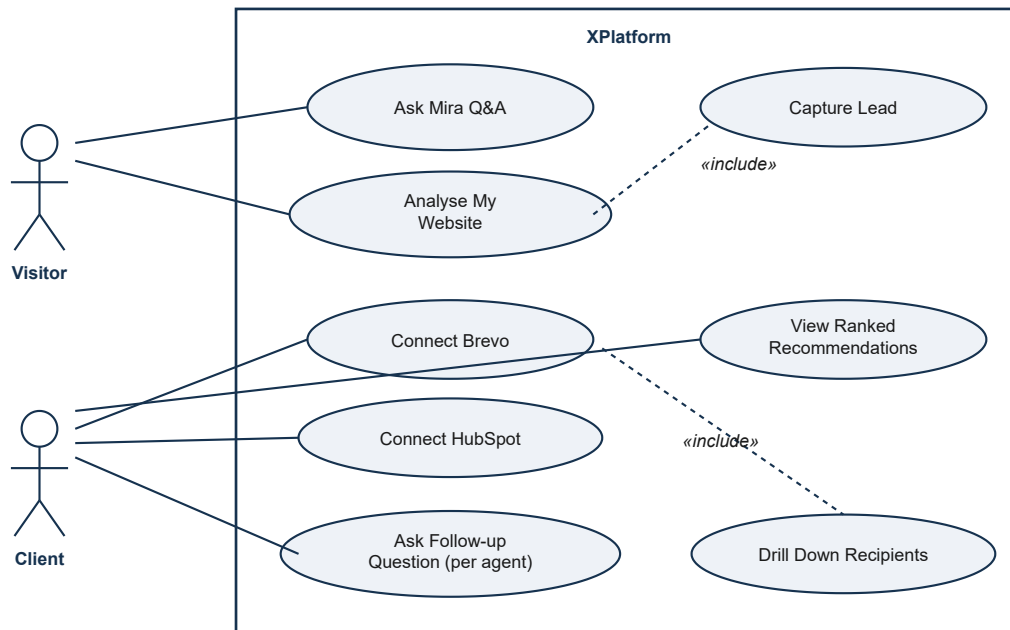


Figure 2.1 XPlatform Use Case Diagram (Ask Mira and Business Helpers / connections)

2.4.2 Use Case Descriptions

Table 2.3 Use Case Descriptions

USE CASE	DETAIL
UC1 — Analyse My Website	Actor: Visitor (unauthenticated). Precondition: None — publicly accessible from the homepage. Main flow: (1) Visitor opens the Ask Mira widget and selects "Analyse my website" or pastes a URL directly. (2) System runs <code>WebsiteAnalyzerService</code> against the single page: title/meta length, heading structure, image alt-text, security headers, mobile signals, link health, robots/sitemap, Open Graph tags, structured-data eligibility. (3) System returns a short chat reply plus a full scored report (0–100, letter grade A–F) below the chat. (4) System records the visitor's name/email/URL as a lead. Alternate flow: Visitor requests a full multi-page crawl or paid-API scoring (e.g. Core Web Vitals) — system declines honestly and links to the Pricing page instead of attempting an unsupported check. Postcondition: A <code>ChatBotUser</code> lead record exists; if the visitor is also logged in as a client elsewhere, the exchange is additionally saved to their account chat history in a separate table.

UC2 — View Ranked Recommendations (Business Helper)	Actor: Client (authenticated, onboarded). Precondition: Client has connected at least a CRM (for deal/contact data); Brevo connection is optional and improves trust/frustration scoring when present. Main flow: (1) Client opens Marketing, Sales, or Customer Retention inside Business Helpers. (2) The agent's current step (e.g. Sales' Prioritise) queries real <code>CrmContact</code> / <code>CrmDeal</code> rows, computes segment/engagement/intent/churn/trust/loyalty scores, and renders a ranked stack. (3) Client reviews the top account and uses a one-click action (Call Now, Launch, etc.) or asks the embedded Ask Mira side-chat a follow-up question from the pre-loaded quick prompts. Postcondition: Client has a prioritised, explainable next action grounded in their own real data.
UC3 — Sync & Drill Down Brevo Engagement	Actor: Client. Precondition: Client has connected a Brevo account. Main flow: (1) Client clicks "Sync Data". (2) System kicks off the <code>BrevoDeliveredRecipient</code> background export/sync job against Brevo's API, with rate-limit-aware retry. (3) Panel header updates to show sync time and campaign count. (4) Client clicks "View" under any of the five metrics (Delivered, Opens, Clicks, Conversions, Unsubscribes) and sees the real recipient list behind that number. Postcondition: Engagement data is current and also available to Business Helpers' trust/frustration scoring for the same contact.

2.5 Context Model

The context model treats XPlatform as a single process and shows every external entity it exchanges data with. Two categories of external actor exist: **people** (Visitor, Client, Master Admin) and **third-party APIs** the system calls out to (Anthropic Claude, OpenAI, Brevo, HubSpot). No new external entity was added by this cycle — Anthropic and OpenAI were already integrated providers; this cycle changed *how* Ask Mira and the Business Helper agents use them (compute-first, narrate-second), not which systems are involved.

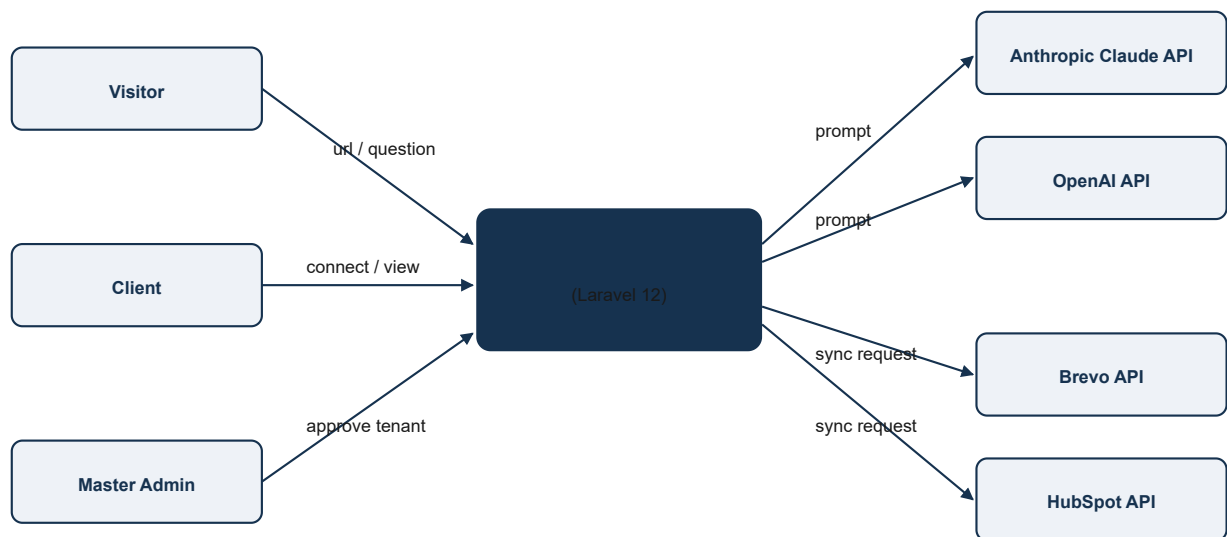


Figure 2.2 Context Model (Level-0 Data Flow)

3. Design

3.1 Architectural Design

XPlatform follows a layered Laravel MVC-plus-services architecture. Controllers stay thin; each meaningful piece of business logic — scoring, ranking, syncing, narrating — lives in a dedicated Service class under `app/Services`, which is the pattern this cycle's work continues rather than departs from (e.g. `MarketingPerformanceService`, `SalesPromptInsightsService`, `WebsiteAnalyzerService`). Background work (Brevo sync) runs as a queued Job so a rate-limited or slow third-party call never blocks a page response.

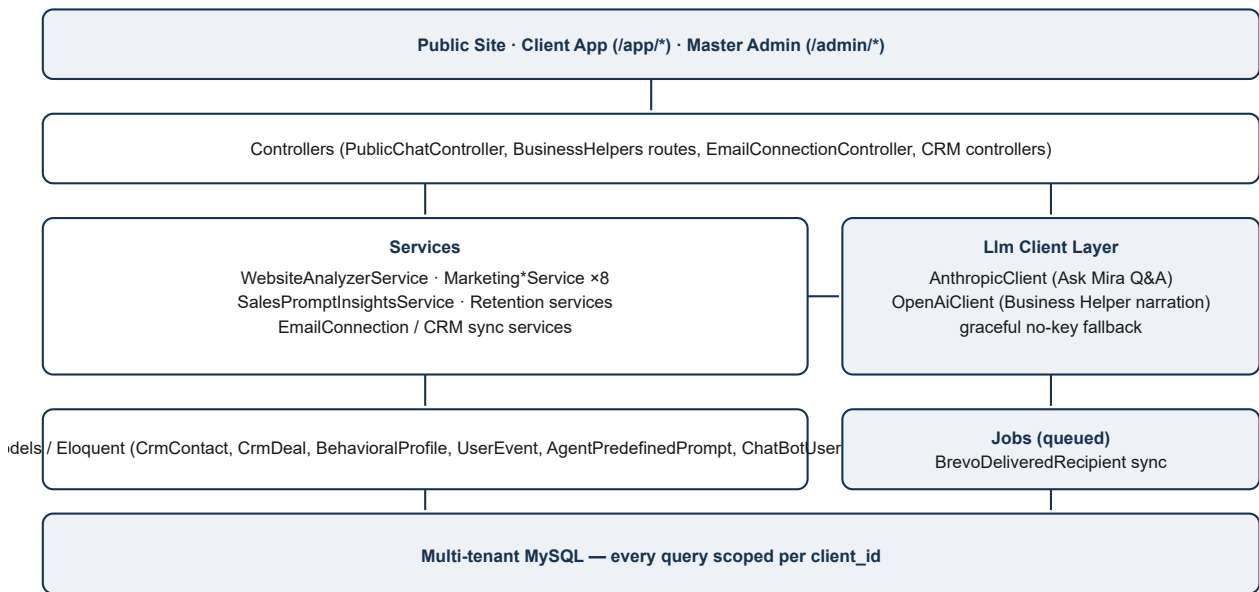


Figure 3.1 XPlatform System Architecture (this cycle's layers highlighted)

3.2 Hardware Specifications

XPlatform is a server-rendered SaaS web application; it has no bespoke hardware requirement, and none of this cycle's features (an AI chat widget, background sync jobs, dashboard rendering) change that. Indicative requirements:

COMPONENT	MINIMUM
Application server	PHP 8.2+, 2 vCPU / 4 GB RAM minimum for the web process; queue worker process runs alongside for the Brevo sync job.
Database server	MySQL 8.0 (or compatible), sized to the client's synced CRM/email volume — this cycle adds no new database engine.
Client device	Any modern desktop or mobile browser (Chrome, Edge, Safari, Firefox) — no plugin or native app required.
Network	Outbound HTTPS access from the application server to the Anthropic, OpenAI, Brevo and HubSpot APIs.

3.3 Database Structure

The entity-relationship diagram below covers the tables this cycle's features actually read and write. `crm_contacts` and `crm_deals` carry no real foreign key to each other — a deal is matched to a contact by "deal name starts with the contact's company name", a heuristic used consistently across every Business Helper service, since the source CRM does not expose a stronger link.

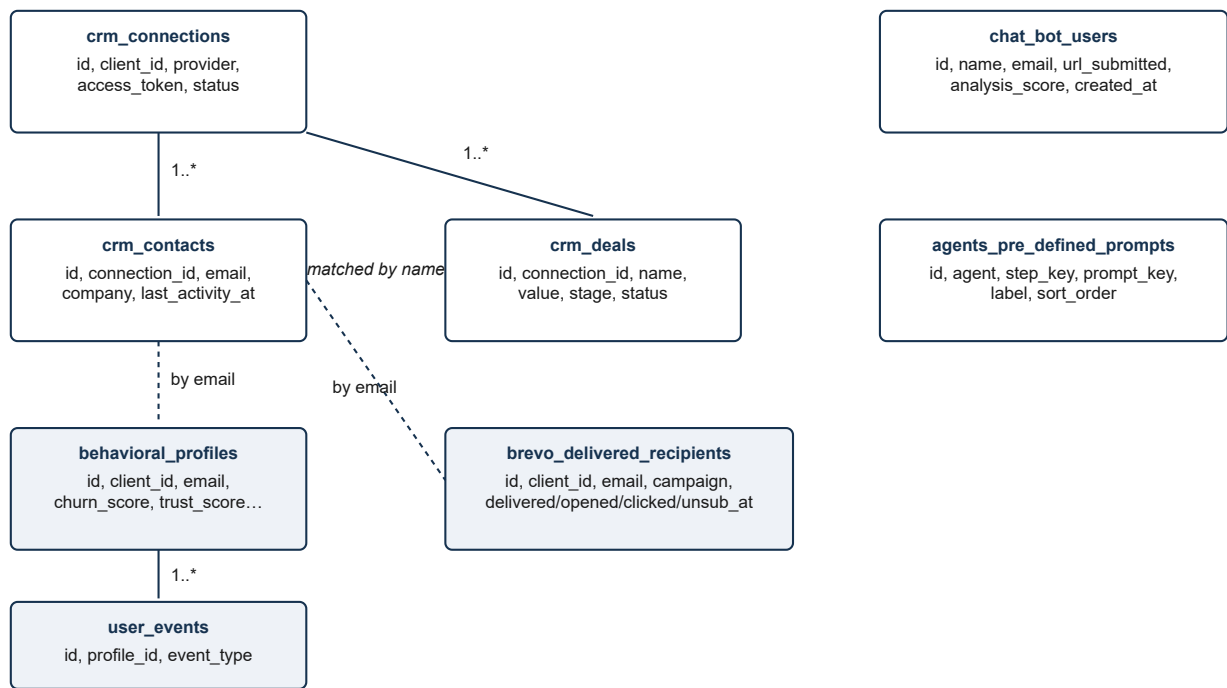


Figure 3.2 Entity-Relationship Diagram (core tables underpinning this cycle's features)

3.4 Interface Design

The Business Helpers workspace is the primary new interface surface: a three-column layout — a left numbered step list, a centre panel that renders the current step's ranked content, and a right-hand embedded Ask Mira chat pre-loaded with step-specific quick prompts. The Brevo and HubSpot panels follow the platform's existing card-and-chart report layout, extended with a recipient drill-down modal for Brevo.

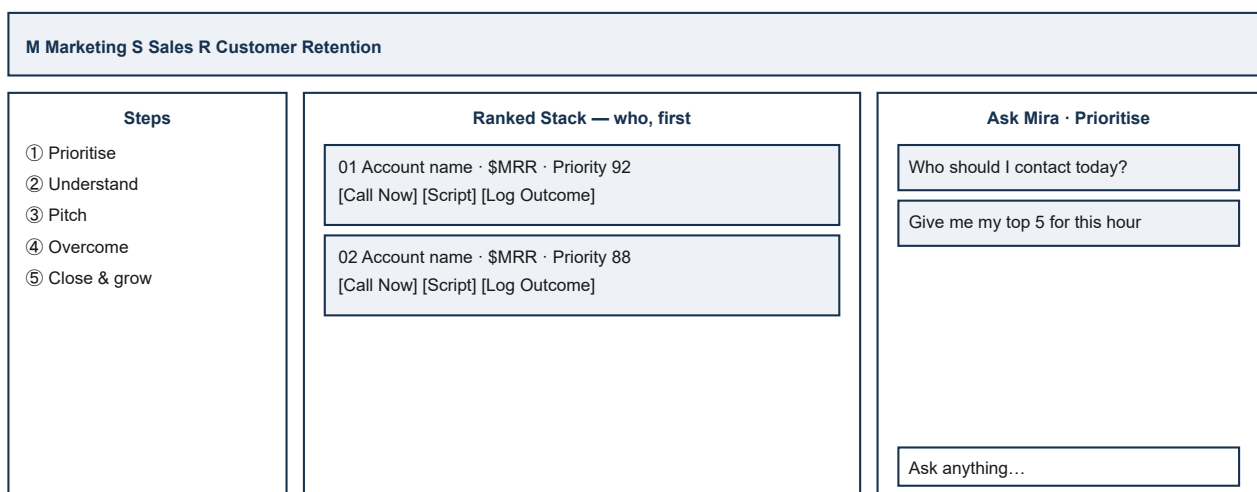


Figure 3.3 Business Helpers Interface Layout (Wireframe)

3.5 Sequence Diagrams

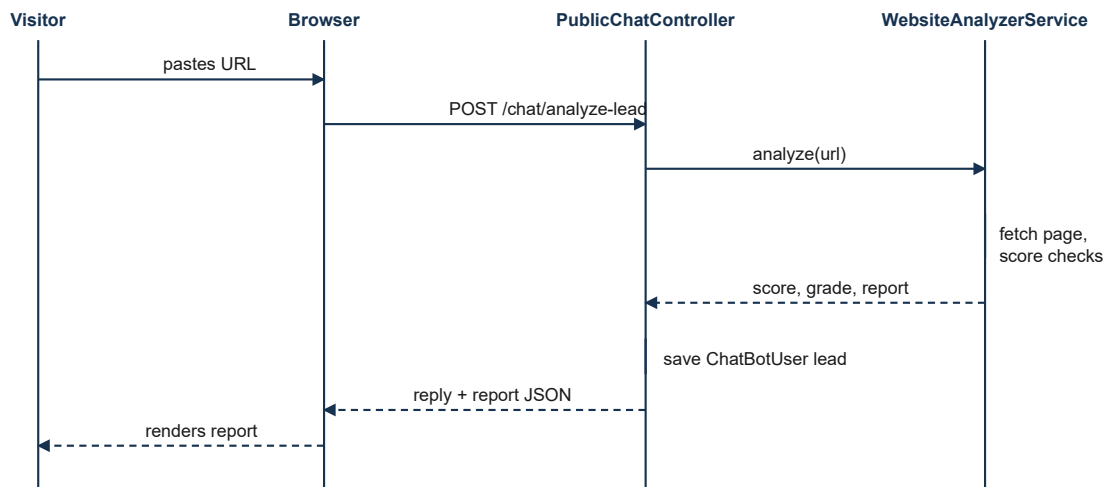


Figure 3.4 Sequence Diagram — Ask Mira Website Analysis

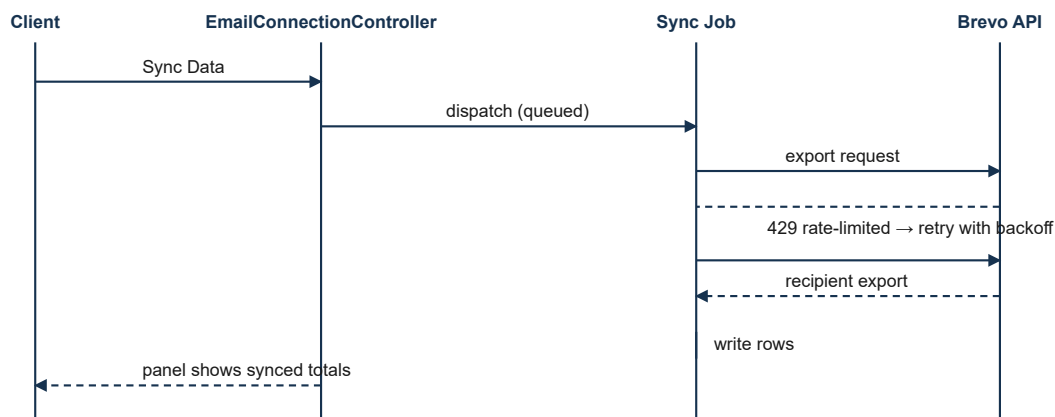


Figure 3.5 Sequence Diagram — Brevo Sync & Recipient Drill-down

3.6 State Diagrams

A client account's lifecycle governs when the features specified above become reachable — Business Helpers, Brevo and HubSpot connections are all gated behind an **Active, Onboarded** state.

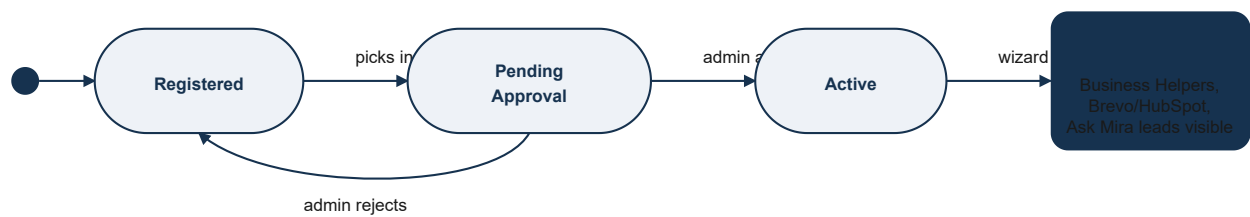


Figure 3.6 Client Account State Diagram

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